

Instructions (shown before students start the test)

Hello competitors! Best of luck on this test :)

Google form for video submission of hydrometer testing: <https://forms.gle/FhY5J4LyZAJ4Tc6T7> (<https://forms.gle/FhY5J4LyZAJ4Tc6T7>) (<https://forms.gle/FhY5J4LyZAJ4Tc6T7>)

You must complete the written exam within 50 minutes of opening the exam, but you may submit the video to the Google form within 3 hours after starting the exam (to accomodate for technical difficulties).

Requirements for the video:

1. Please limit the total length of the video to under 10 minutes. Please try to film all in one continuous shot and avoid multiple videos if possible.
2. Both solutions and the active testing performed by students and their hydrometer must be included in the video.
3. Clearly show a close-up of the hydrometer.
4. Clearly show the students verbally declaring the final values from their hydrometer readings of both solutions that they wish to submit for grading.

Introduction (shown after students start the test)

The code word for the video is: cotton candy

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1. (1.00 pts) Which of the following incorrectly describes a disaccharide sugar?

- ☐ A) Sucrose is a combination of glucose and fructose.
- ☐ B) Lactose is a combination of glucose and galactose.
- ☐ C) Maltose is a combination of glucose and glucose with a 1,4 glycosidic bond.
- ☐ D) Chitobiose is a combination of glucose and glucose with a 1,1 glycosidic bond.

2. (1.00 pts)

Phenylketonuria (PKU) is characterized by the lack of enzyme ____, which allows for the breakdown of phenylalanine, an amino acid found in the sweetener ____.

- ☐ A) Phenylalanine dehydrogenase; advantame
- ☐ B) Phenylalanine dehydrogenase; aspartame
- ☐ C) Phenylalanine hydroxylase; aspartame
- ☐ D) Phenylalanine hydroxylase; advantame

3. (1.00 pts) Which of the following is an incorrect statement about canning?

- ☐ A) One downside of canning is a loss of nutrients during the process.
- ☐ B) Canning involves first heating and then packaging food in a vacuumed container.
- ☐ C) Canning is a type of food preservation.
- ☐ D) Parts of the food that are closer to the heat source are typically overheated.

4. (1.00 pts) Benedict's Test involves a reaction between a reducing sugar's ____ and the ____ in Benedict's solution.

- ☐ A) Aldehyde; copper sulfate
- ☐ B) Aldehyde; copper sulfide
- ☐ C) Ketone; copper sulfate
- ☐ D) Ketone; copper sulfide

5. (1.00 pts) The Maillard reaction occurs best at which temperature range?

- ☐ A) 160 to 185 °C
- ☐ B) 120 to 145 °C
- ☐ C) 140 to 165 °C
- ☐ D) 100 to 125 °C

6. (1.00 pts) Which of the following is an incorrect step of the Krebs cycle?

- ☐ A) Malate dehydrogenase turns malate into oxaloacetate
- ☐ B) Aconitase turns citrate into isocitrate
- ☐ C) Succinate dehydrogenase turns succinate into fumarate
- ☐ D) Isocitrate dehydrogenase turns isocitrate into succinyl-CoA

7. (1.00 pts) Which of the following is not a product of the Krebs cycle?

- ☐ A) CO₂
- ☐ B) O₂
- ☐ C) NADH
- ☐ D) FADH₂

8. (1.00 pts) In which of the sugar stages does caramelization occur?

- ☐ A) Brown liquid stage
- ☐ B) Hard crack stage
- ☐ C) Firm ball stage
- ☐ D) Burnt Liquid stage

9. (1.00 pts) Which of the following is a non-essential biological macromolecule?

- ☐ A) Omega-3 fatty acids
- ☐ B) Galactose
- ☐ C) Cysteine
- ☐ D) Isoleucine

10. (1.00 pts) Which class of enzymes is only found in the stomachs of ruminant mammals?

- ☐ A) Amylase
- ☐ B) Oxireductase
- ☐ C) Rennet
- ☐ D) Protease

11. (1.00 pts) What is the most commonly used font in nutrition labels?

- ☐ A) Helvetica
- ☐ B) Calibri
- ☐ C) Arial
- ☐ D) Times New Roman

12. (1.00 pts) Which allergen is currently not required to be labeled?

- ☐ A) Shellfish
- ☐ B) Soybeans
- ☐ C) Tree nuts
- ☐ D) Sesame

13. (1.00 pts) Which disease is characterized by a gluten allergy?

- ☐ A) Peptic ulcer disease
- ☐ B) Celiac disease
- ☐ C) Crohn's disease
- ☐ D) Irritable bowel syndrome

14. (1.00 pts) Which of the following is false regarding leavening agents?

- ☐ A) These include anything that can produce a gas that will lighten and expand dough or batter.
- ☐ B) Salt is a leavening agent, as it controls the rate of yeast fermentation, which impacts how much dough can rise.
- ☐ C) During leavening, gas gets trapped in a matrix of gluten and starch.
- ☐ D) Baking soda is a good leavening agent, but only works well in the presence of moisture and an acidic substance.

15. (1.00 pts)

An acidified food (AF) is a low-acid food to which acid(s) or acid food(s) are added and which has a finished equilibrium pH of ____ or below and a water activity (aw) greater than ____.

- ☐ A) 4.6; 0.85
- ☐ B) 4.6; 0.75
- ☐ C) 3.6; 0.75
- ☐ D) 3.6; 0.95

16. (1.00 pts) Which of the following is the correct definition of water activity?

- ☐ A) The difference between the partial vapor pressure of water in a food substance to the partial vapor pressure of distilled water at the same temperature.
- ☐ B) The ratio of the partial vapor pressure of distilled water to the partial vapor pressure of water in a food substance at the same temperature.
- ☐ C) The ratio of the partial vapor pressure of water in a food substance to the partial vapor pressure of distilled water at the same temperature.
- ☐ D) The ratio of the partial vapor pressure of water in a food substance to the partial vapor pressure of deionized water at the same temperature.

17. (1.00 pts) Which of the following is an enzyme that breaks down proteins into amino acids and is usually found in the small intestines?

- ☐ A) Kinase

- ☐ B) Chymosin
- ☐ C) Pepsin
- ☐ D) Trypsin

18. (1.00 pts) Which of the following is true regarding the brown bag test?

- ☐ A) If the bag is more transparent, that means there are less proteins
- ☐ B) If the bag is less transparent, that means there are less lipids
- ☐ C) If the bag is more transparent, that means there are less lipids
- ☐ D) If the bag is less transparent, that means there are more proteins

19. (1.00 pts) Which of the following is not in Biuret's solution?

- ☐ A) Copper(II) sulfate
- ☐ B) Sodium hydroxide
- ☐ C) Sodium citrate
- ☐ D) Potassium sodium tartrate

20. (1.00 pts) Which temperature is in the "Danger Zone" for food?

- ☐ A) 138° C
- ☐ B) 138° F
- ☐ C) 32° F
- ☐ D) 220° F

21. (2.00 pts) Name the 6 high-intensity sweeteners that are currently approved by the FDA

22. (2.00 pts) Describe the reaction and/or complex formation that occurs in Biuret's test.

23. (2.50 pts) What do each of the following acronyms stand for?

- a) NAD⁺
- b) FAD
- c) ADP
- d) GTP
- e) CoA

24. (5.00 pts) List at least FIVE changes that occurred to nutrition labels in January of 2021.

25. (8.00 pts) Label the various enzymes A-H in the diagram below.

26. (3.00 pts)

When regular and diet soda are both put in water, regular soda sinks and diet soda floats. However, a packet of regular mayonnaise floats higher than a packet of light mayonnaise. Explain both of these phenomena.

27. (3.00 pts)

Lyophilization is a method of drying foods for the purpose of food preservation. Describe the process of lyophilization. Specifically, distinguish between primary and secondary drying.

28. (2.00 pts) Describe the difference between nucleation and crystallization.

Use the food label below to answer questions 29-32:

29. (1.00 pts) What are “added sugars”?

30. (2.00 pts) How many mg of potassium are in 100g of this food? Show units.

31. (2.00 pts) What is the structural difference between saturated and trans fats? Which one is less healthy?

32. (2.00 pts) Based on this nutrition label, how many mg of calcium are in one container of this food if the recommended daily value for calcium is 260mg? Show units.

33. (2.00 pts) What is the difference between moisture content and water activity. Which one is more useful for the purposes of food preservation, and why?

34. (2.00 pts) List all of the fat-soluble essential vitamins.

35. (8.00 pts) Name the sugar stage, describe the properties of sugar, and give the percentage of sugar in water at the following temperatures.

- a) 237° F
- b) 305° F
- c) 338° F
- d) 354° F

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Congrats on finishing, and thank you for participating! Best of luck with your other events :)